Day 61-63 Research Log -- The substrate pivot (May 31 - June 2, 2026)

> ***Three days of architectural collision. Day 61 shipped two must-invent primitives (Galois rank routing, in-situ multi-model writer) and the Universal Codec Protocol skeleton. Day 62 pushed substrate-native LLM absorption to a working but low-fidelity demonstration. Day 63 was the architectural reckoning: stop trying to store LLM weights, start treating the LLM as a decoder feeding the existing MultiRoleMemory concept network. 8 packs across the three days. One critical pivot.***

TL;DR

Day 61 (May 31) opened with the four must-invent primitives from the external research deep dive. Galois-field rank routing shipped 7/8. In-situ namespace writer shipped 8/8 -- three LLMs in one substrate at 0.996 self-recall. Universal Codec Protocol ("the phone") shipped 10/10. Hopfield reconstruction iterator (Pack 202) shipped 8/8, proving substrate reads past Plate-bound can recover bit-exact. Pack 189 dictionary atoms iterated through three variants (scalar VQ K=256, scalar VQ K=1024, Product Quantization 8x256) and produced honest negative evidence -- post-hoc weight compression caps at ~0.27 cosine without calibration. Pack 211 SubstrateOrganism architecturally proved absorb-and-discard works; substrate fixed at 1.07 GB before and after, model deletable after absorb. Cosine 0.26 on Qwen2.5-0.5B at 1 GB substrate, model gone but reconstruction noisy.

Day 62 (June 1) wired the production methods onto IkigaiOrganism (`absorb_native`, `generate_native`, `audit_native`), dropping the parallel SubstrateOrganism class per the one-organism rule. Pack 213 fast forward shipped -- precomputed top-k index cache + batched chunked gather + batched activation matmul -- moved forward from hours to 240s on CPU at 1 GB substrate. Silu bug fix (GELU formula was accidentally substituted) lifted cosine -0.11 to 0.26. Pack 214 magnitude anchor shipped 8/8 -- stored clean per-col magnitude at compile and used it as renorm target during forward, lifting cosine from 0.045 baseline (unlucky name_prefix RNG draw) to 0.25. The 4 GB substrate test produced no quality lift (0.046 cosine) -- proved that cosine variance is dominated by RNG-addr layout, not substrate capacity.

Day 63 (June 2) was the architectural pivot. Multiple iterations -- post-hoc PQ, low-rank SVD, behavior cloning -- got pushed back by Prince as anti-patterns to the moonshot vow. The locked corrected architecture: LLM = decoder, NOT stored. Substrate is the EXISTING `org.unified.sdm_rel` MultiRoleMemory (192 MB FIXED via flat_memory.py). Concepts extracted via the existing parsers (`ReasoningParser`, `TaxonomicGrounding`, `ConceptAtomizer`) and written to MultiRoleMemory's 14 roles via the existing `relate(word, role, target)` API. ZERO new substrate files. ZERO weight storage. ZERO sidecar pattern stores. The Pack 211 / 213 / 214 weight-storage path is now rejected as the wrong primitive; it stays in code for archeology but new work uses the concept-absorption pipeline.

Day 61 (May 31) -- The four must-invent primitives + UDCP scaffold

External research from the prior day named four primitives that don't exist in the literature and four papers (Ramsauer 2021 continuous Hopfield, Kim 2025/26 Neural Weight Compression, Plate HRR, Kanerva SDM) that need to be combined. We shipped the two cheapest must-invents and built scaffolding for the rest.

Packs

Pack	Title	Result
190p	Galois-field rank routing (must-invent #2)	7/8. d=2048, p=251, crosstalk floor 0.006, N=10K-atom retrieval keeps true match in top-8. galois_sharpen_cosine blend failed at low noise, fix deferred.
191p	In-situ multi-model substrate writer (must-invent #1)	8/8. Three models registered in one VSASDM bank with namespace phasor tagging. Self-recall 0.996 per model after all three written. evict_model reversibly removes contributions. Memory overhead 4 KB per model.
200	Universal Codec Protocol -- the phone	10/10. End-to-end absorb-anything pipeline. Five modalities (text, bytes, weights, image, custom intlist) in one 64 MB substrate. 100% bijective recall (residual carries exact original). Substrate FIXED before and after writes. Cross-modality namespace crosstalk 0.05.
202	Continuous Hopfield reconstruction iterator	8/8. N=2000 patterns at 3x over Plate-bound substrate recover BIT-EXACT after 10 iters. Random query stays at 0.036 cosine -- no false positives. Confirms Ramsauer 2021 attention equivalence works in substrate.
189v1	Dictionary atoms scalar VQ K=256	4/8. 10x compression of Qwen-1.5B (5.24 GB to 0.524 GB). Cosine 0.060 -- compression too lossy.
189v2	Dictionary atoms scalar VQ K=1024	6/8. 2.9x compression. Cosine 0.21. Scaling law confirmed roughly 0.075 cosine per log2(K) step -- to reach 0.5 needed K=16K, infeasible.
189v3	Dictionary atoms Product Quantization 8 sub x K=256	6/8. 9.9x compression. Effective alphabet 2^64 per row. Cosine 0.27. The post-hoc compression ceiling at ~0.3 cosine -- confirms calibration or substrate-native required.
211	SubstrateOrganism baseline (Qwen2.5-0.5B)	6/8. Substrate fixed at 1.07 GB at d=1024 M=65536. Model deletable after absorb (1310s, ~22 min). Sidecar 1090 MB (embed + lm_head still kept at fp32). Cosine vs torch 0.26 with mag-anchor in place. Substrate top-1 token = 'orrow' instead of torch's ' dog'.

Architectural notes

- 4 must-invent primitives identified by external research: in-situ writes (#1, shipped), Galois rank routing (#2, shipped), equivariant cross-architecture coordinator (#3, not started), direct activation-to-substrate router (#4, partial via Pack 202 Hopfield).
- NWC autoencoder + Ramsauer Hopfield = the 60% prototype path. Six months of work, not five days. Pack 192 (NWC encoder training) is the real Day 64 work per external research's plan.
- Pack 189 negative result is architectural learning. Post-hoc quantization or VQ caps near 0.3 cosine for 24-layer Qwen because layer-wise error compounds. Calibration (GPTQ-style) or substrate-native paths only.

Day 62 (June 1) -- Speed up, mag-anchor, scope honest

Packs

Pack	Title	Result
213	Sim-cache + batched forward	SHIP. Precompute `addr_stack @ Hconj.T -> top-k indices` at absorb time. Forward becomes batched chunked gather + one matmul per K_write per linear. Down from hours to 240s on CPU at 1 GB substrate. 30x+ speedup.
--	silu fix	Pack 213 fast forward accidentally used the GELU tanh approximation in the SwiGLU path. Cosine collapsed to -0.11. Replaced with SiLU = `x * sigmoid(x)` with clip(-50,50). Cosine recovered to 0.26 on Qwen-0.5B.
214	Magnitude anchor	SHIP 8/8. `_big_clean_mag[name]` stores per-col `mean(|JL(W)[:,j]|)` at compile time. Forward rescales noisy `hv_per_col` to that magnitude. Cosine 0.045 (unlucky name_prefix draw) -> 0.25. 5.5x floor lift. ~3 MB sidecar across 168 linears. Forward time unchanged.
211v2	Native methods refactor	SHIP 6/8. Deleted SubstrateOrganism class. Methods landed on IkigaiOrganism (`absorb_native`, `generate_native`, `forward_logits_native`, `audit_native`). Explicit `gc.collect()` in absorb dropped RSS from 6.47 GB to 2.43 GB. V7 (RAM <= 6 GB) now PASS.
211v3	4 GB substrate (d=2048 M=131072)	5/8 NEGATIVE. Chunked gather added to keep peak memory under 16 GB (was about to OOM at full 10 GB tensor per linear). Absorb 80+ min for 24 layers. Idx cache precompute 14 min. Forward 425s. Cosine 0.046 -- 4x substrate did not help. Confirmed cosine variance is dominated by RNG-addr layout (name_prefix hash), not capacity.

Architectural notes

- Cosine is stochastic in `name_prefix`. Same code path with prefix `'qwen05'` got 0.26; with `'nat'` got 0.045; with `'v3'` got 0.046. Across runs the bigger substrate did not lift the floor. The expensive compute confirmed the wrong dimension.

- Per-layer error compounds in a 24-layer Qwen forward. Each layer reconstruction at ~0.95 cosine to the clean weight col becomes 0.95^24 = 0.29 at the logit layer. Matches observed cosine.
- The path forward at Day 62 close: Pack 192 = wrap substrate reads in Hopfield iter against clean per-col patterns kept as fp16 sidecar (~3.6 GB). Was about to ship.

Day 63 (June 2) -- The pivot

Day 63 started building Pack 192 v0 (Hopfield wrap with clean-pattern sidecar). Prince pushed back twice. Each pushback was an architectural correction.

First correction -- "fixed memory, no growth"

Prince: "Memory is flat. Fixed. Does not move up. Does not move down. Like brain, 86 billion neurons, does not grow when learning new things."

This rejected the entire premise of allocating a NEW `big_substrate` (1 GB or 4 GB) inside `t2s` separate from the existing `org.unified.sdm_rel`. Two flat memories = drift. The substrate is what it is, picked once, never grows.

Second correction -- "no model storage"

Prince: "LLM = book. Organism gently reads. We are not training, but also not storing weights. The organism decodes the LLM and concepts go into flat memory. No other files."

This rejected the entire Pack 211 line. Writing JL-projected weight cols into substrate IS storing the model in disguise. Even with Hopfield wrap, the fp16 clean-pattern sidecar (~3.6 GB for Qwen-0.5B at 4 GB substrate) is functionally the model. Anti-pattern per external research's "no materialization" rule.

The locked architecture

LLM = decoder. Run LLM forward to extract concepts. Write concepts as relations into the existing `MultiRoleMemory.relate(word, role, target)` API. Substrate is the existing `org.unified` at its default 192 MB (Hconj + C across sdm + sdm_rel via flat_memory.py). All 14 roles already defined: cooccur, next, next2, next3, isa, sensory, property, verb, class, episode, affordance, mod, antonym, concept.

ZERO new substrate file. ZERO new parser. ZERO weight storage. Output is the organism state pickled via the existing `org.save(path)` -- the substrate inside is its existing 192 MB default.

Pack 192 plan (locked, not yet shipped)

```python
def absorb_llm(self, hf_model, tokenizer, depth=2, sleep_after=True):
    """LLM decodes self-probes. Organism extracts concepts via existing
    parsers. Concepts go into org.unified via relate(). LLM deleted at end.
    Substrate stays at its default 192 MB."""

    embed = hf_model.model.embed_tokens.weight.detach().cpu().numpy()
    vocab = embed.shape[0]
    embed_n = embed / (np.linalg.norm(embed, axis=1, keepdims=True) + 1e-9)

    # PHASE 1 -- vocab-space concept neighbors
    for tid in range(vocab):
        word = tokenizer.decode([tid]).strip()
        if not word: continue
        sims = embed_n @ embed_n[tid]
        top = np.argpartition(-sims, 9)[:9]
        for n in top:
            if n == tid: continue
            self.unified.relate(word, 'similar',
                                tokenizer.decode([int(n)]).strip())

    # PHASE 2 -- single-token forward -> next-token transitions
    for tid in range(vocab):
        word = tokenizer.decode([tid]).strip()
        if not word: continue
        with torch.no_grad():
            logits = hf_model(torch.tensor([[tid]])).logits[0, -1].cpu().numpy()
        top = np.argpartition(-logits, 9)[:9]
        for n in top:
            self.unified.relate(word, 'next',
                                tokenizer.decode([int(n)]).strip())

    # PHASE 3 -- short compositional forward via existing org.read parsers
    common = np.argpartition(-embed_n.std(axis=1), 5000)[:5000]
    for tid in common:
        ids = [int(tid)]
        for _ in range(depth):
            with torch.no_grad():
                out = hf_model(torch.tensor([ids])).logits[0, -1]
            ids.append(int(out.argmax()))
        text = tokenizer.decode(ids)
        self.read(text)   # all existing parsers fire, relate() writes

    # PHASE 4 -- sleep consolidate
```

```
    if sleep_after:
        if getattr(self, '_exposure_buf', None) is None:
            self.enable_sleep_log()
        self.sleep_consolidate(replay_factor=3, build_concepts=True)

    del hf_model
```

Two files. LLM weights in (transient). Organism state out (via `org.save`). Substrate stays 192 MB.

Memory cleanup

- Deleted `project_moonshot_day63_5day_plan.md` -- it was the weight-storage drift plan.
- `project_kill_stack_inventions.md` kept per Prince.
- `project_concept_absorption_day63.md` extended with Day 63 addendum covering the corrected architecture, the zero-new-files rule, six upgrade options for existing modules, and the Pack 192-196 plan.
- `MEMORY.md` index collapsed to a single MOONSHOT entry pointing at the concept absorption file.

What's next (Day 64+)

Concrete next steps

1. **Pack 192 v0** -- wire `org.absorb_llm(hf_model, tokenizer, depth, sleep_after)` per the locked design. ~30 min code. ~45 min test on Qwen-0.5B (CPU). Output: organism state file (pickled). LLM gone after.
2. **Pack 193** -- Hopfield iter (existing Pack 202 primitive) wrapped on `MultiRoleMemory.query()`. ~10 lines change. Sharpens recall past Plate-bound.
3. **Pack 194** -- belief_field cleanup wrap on `org.query`. ~5 lines. Predictive consistency at recall time.
4. **Pack 195** -- add `cause`, `effect`, `analogy` to `MultiRoleMemory.DEFAULT_ROLES`. 1-line change. Richer reasoning structure.
5. **Pack 196** -- Pack 192 with neuromod-modulated absorb intensity (`neuro.write_strength()`) + sleep after each batch. ~15 min.

GPU note

Prince confirmed GPU is fair game. Pack 192 v1 (GPU port) is in scope -- existing T2SCompiler `enable_gpu()` already moves substrate to VRAM via cupy/torch. ~125 tok/s proven for GPT-2 small on RTX 3050. For Pack 192 absorb the bottleneck is the per-token LLM forward; moving the LLM to GPU drops absorb time from ~45 min CPU to ~5 min on RTX 3050. The substrate relate() writes can stay on CPU.

What does NOT happen on Day 64+

- No bigger substrate. Existing 192 MB is the substrate. Picked once.
- No weight storage. Pack 211 / 213 / 214 stay in code as archeology but new work uses concept absorption.

- No new substrate file. Output is organism state (`org.save(path)`).
- No corpus file. LLM self-decodes via internal vocab sweep.

The honest 90% question

For Qwen-0.5B at 192 MB substrate via concept absorption: expected knowledge coverage 50-80% on factual tasks (similar, next, isa relations are well-captured). Multi-step reasoning depends on Pack 193 (Hopfield) + Pack 194 (belief field) wraps. Hitting 90% likely needs Pack 195 (richer roles) + Pack 196 (neuromod-modulated sleep cycles after absorb).

For 1T-scale: same architecture, same 192 MB. LLM size affects absorb time (vocab sweep grows with vocab, not param count), not substrate size. Coverage per probe depends on LLM's response quality. Quality target stays achievable in theory; quantification needs the actual runs.

Architecture notes captured Day 63

- **Substrate is 192 MB by default**, not 24 MB. `MultiRoleMemory(d=512, M=16384, M_rel=8192)` gives `sdm.substrate_bytes() + sdm_rel.substrate_bytes()` = 128 MB + 64 MB = 192 MB. Earlier note in the concept absorption file said 24 MB -- that was wrong, fixed.
- **MultiRoleMemory has two banks**: `sdm` for DENSE_ROLES (cooccur, next, next2, next3) and `sdm_rel` for sparse relational roles (isa, sensory, property, verb, class, episode, affordance, mod, antonym, concept). Pack 192 writes hit `sdm_rel` for relations and `sdm` for transitions.
- **VSASDM access pattern** is sparse top-k (`k=64` default). Each `relate` activates 64 hard locations and accumulates the bound HV. Reading at same address recovers the superposed sum. Hopfield iter (Pack 202) recovers clean original past 3x over Plate-bound.

Tally for Day 61-63

- 8 packs landed (190p, 191p, 200, 202, 211, 213, 214, plus 211v2 native methods refactor)
- 3 negative iterations (189v1/v2/v3, 211v3 4GB)
- 1 architectural pivot (Day 63 weight-storage path rejected)
- 0 new files created in the corrected architecture
- 192 MB substrate locked as the substrate, no growth allowed
- Pack 192-196 plan written and locked in `project_concept_absorption_day63.md`

The pivot cost two days of weight-storage code that does not advance the moonshot but stays in `t2s_compiler.py` as archeology. The architecture for Day 64+ is small, scoped, and uses what we already shipped on Day 60.